Empirical–generated learning-curve comparisons
Full observed source ranges

A BRCA miRNA-seq
TC · cVAE 1:5
No AE head
Mean source ARI: 0.0055

B HNSC miRNA-seq
raw · cVAE 1:50
No AE head
Mean source ARI: 0.0058

C HNSC miRNA-seq
TC · cVAE 1:50
No AE head
Mean source ARI: 0.0080

D COAD RNA-seq
TC · cVAE 1:50
AE head (2 iterations)
Mean source ARI: 0.00014

	A	B	C	D
LR AUC	B +0.053 D 0.053	B +0.036 D 0.036	B +0.144 D 0.144	B +0.124 D 0.124
SVM AUC	B +0.028 D 0.034	B -0.008 D 0.023	B +0.136 D 0.136	B +0.047 D 0.117
KNN AUC	B +0.008 D 0.016	B -0.019 D 0.025	B +0.166 D 0.166	B +0.075 D 0.092
RF AUC	B +0.007 D 0.013	B -0.022 D 0.031	B +0.123 D 0.123	B +0.071 D 0.071
XGB AUC	B +0.019 D 0.024	B -0.032 D 0.039	B +0.093 D 0.093	B +0.072 D 0.079

AUC: 0.00, 0.25, 0.50, 0.75, 1.00
Candidate subset size: 300, 600, 900 (A); 250, 500, 750, 1000 (B–D)

Empirical
Generated

Points: observed means ±1 draw SD. Lines/bands: archived fit / pointwise 95% fitted-mean interval.
Pale blue/orange: empirical-only/generated-only support. Dotted boundary: upper shared limit.
B: signed average gap; D: absolute average gap. Both use shared observed means. ARI: mean of five fidelity draws.

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LR Macro-F1: B +0.073 D 0.073 | B +0.007 D 0.019 | B +0.146 D 0.146 | B +0.119 D 0.119
SVM Macro-F1: B -0.002 D 0.018 | B +0.005 D 0.015 | B +0.147 D 0.147 | B +0.077 D 0.080
KNN Macro-F1: B +0.001 D 0.009 | B +0.001 D 0.020 | B +0.159 D 0.159 | B +0.069 D 0.079
RF Macro-F1: B -0.006 D 0.009 | B -0.007 D 0.017 | B +0.134 D 0.134 | B +0.054 D 0.054
XGB Macro-F1: B +0.005 D 0.016 | B -0.033 D 0.040 | B +0.096 D 0.096 | B +0.058 D 0.063

Candidate subset size

Empirical Generated

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AE head (2 iterations)
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	A	B	C	D
LR Accuracy	B +0.057 D 0.057	B +0.008 D 0.017	B +0.139 D 0.139	B +0.123 D 0.123
SVM Accuracy	B -0.001 D 0.013	B -0.003 D 0.010	B +0.117 D 0.117	B +0.077 D 0.078
KNN Accuracy	B -0.008 D 0.009	B -0.008 D 0.016	B +0.129 D 0.129	B +0.069 D 0.075
RF Accuracy	B -0.008 D 0.009	B -0.010 D 0.016	B +0.113 D 0.113	B +0.056 D 0.056
XGB Accuracy	B +0.003 D 0.009	B -0.027 D 0.032	B +0.089 D 0.089	B +0.060 D 0.062

Accuracy: 0.00, 0.25, 0.50, 0.75, 1.00
Candidate subset size: 300, 600, 900 (A); 250, 500, 750, 1000 (B, C, D)

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Generated

Points: observed means ±1 draw SD. Lines/bands: archived fit / pointwise 95% fitted-mean interval.
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